SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 20%

Code of Conduct:

*I **K.L DINESH ESHWAR (192425027)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Scenario-Based Questions

1. **Edge Noise Problem**

Edges detected are affected by noise.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in edge detection.
(c) Apply a suitable solution.
(d) Justify your method logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The image contains **noise**, causing the detected edges to be irregular, broken, or contain many false edges. This makes it difficult to accurately identify the true boundaries of objects.

(b) Key Challenges:

The major challenges include **image noise**, false edge detection, weak or discontinuous edges, blurred boundaries, and reduced accuracy in object detection and segmentation.

(c) Suitable Solution:

The most suitable solution is to apply **Gaussian Filtering** for noise removal, followed by **Canny Edge Detection**. The Gaussian filter smooths the image, while the Canny algorithm detects accurate and continuous edges.

(d) Justification:

This approach is effective because **Gaussian Filtering** removes random noise without excessively blurring the image, and **Canny Edge Detection** uses gradient analysis and non-maximum suppression to produce thin, continuous, and accurate edges. It also reduces false edge detection through double thresholding and edge tracking.

(e) Structured Explanation:

When edge detection is affected by noise, the best approach is to **first remove noise using a Gaussian Filter and then apply Canny Edge Detection**. This sequence minimizes false edges, preserves important object boundaries, and produces clear and continuous edges, improving the accuracy of image analysis.

2. Region Merging Requirement

An image contains fragmented regions that need merging.

(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in region segmentation.
(c) Apply an appropriate technique.
(d) Justify your approach logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The image contains **fragmented regions**, where a single object is divided into several small segments after segmentation. These fragmented regions need to be merged to represent the complete object accurately.

(b) Key Challenges:

The major challenges include **over-segmentation**, image noise, small disconnected regions, weak boundaries, and incorrect object representation. These issues reduce segmentation accuracy.

(c) Appropriate Technique:

The most suitable technique is **Region Merging** (Region-Based Segmentation) using **Morphological Closing** to combine adjacent regions with similar properties into a single region.

(d) Justification:

Region Merging combines neighboring regions based on similarity in intensity, texture, or color. Morphological Closing fills small gaps and connects fragmented regions, resulting in complete and accurate object segmentation while reducing over-segmentation.

(e) Structured Explanation:

When an image contains fragmented regions, **Region Merging with Morphological Closing** is the preferred approach. It joins similar neighboring regions, removes small gaps, preserves object shape, and improves segmentation accuracy, making the image suitable for reliable analysis.

3. **Over-Segmentation**

An image is divided into too many regions after segmentation.

(a) Interpret the scenario and explain over-segmentation.
(b) Identify key issues affecting segmentation.
(c) Apply a suitable correction method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The image is divided into **too many small regions** after segmentation. This is called **over-segmentation**, where a single object is unnecessarily split into multiple segments, making analysis difficult.

(b) Key Issues:

The major issues include **excessive region splitting**, image noise, weak boundaries, incorrect object representation, and reduced segmentation accuracy.

(c) Suitable Correction Method:

The most suitable method is **Region Merging** (or **Morphological Closing**) to combine adjacent regions with similar intensity or texture into larger, meaningful regions.

(d) Justification:

Region Merging reduces unnecessary small segments by combining similar neighboring regions while preserving object boundaries. It produces complete objects and improves segmentation accuracy better than repeating the segmentation process.

(e) Structured Explanation:

When an image suffers from **over-segmentation**, **Region Merging** is the preferred correction technique. It combines fragmented regions, reduces unnecessary segments, preserves object shape, and produces accurate and meaningful segmentation for further image analysis.

4. Under-Segmentation

Important regions are merged incorrectly during segmentation.
(a) Interpret the scenario and explain under-segmentation.
(b) Identify key issues affecting separation.
(c) Apply a suitable solution.
(d) Justify your approach logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

Under-segmentation occurs when two or more different objects or regions are **incorrectly merged into a single segment** during the segmentation process. This usually happens when the objects have similar intensity values, weak boundaries, or are touching each other. As a result, the segmentation algorithm fails to distinguish individual objects, making further image analysis inaccurate.

(b) Key Issues Affecting Separation:

The major challenges include **similar intensity values**, weak or blurred boundaries, overlapping or touching objects, image noise, and poor contrast. These factors prevent the algorithm from correctly identifying the boundaries between adjacent objects, leading to incorrect segmentation and reduced object recognition accuracy.

(c) Suitable Solution:

The most appropriate solution is **Marker-Controlled Watershed Segmentation**. Before applying the watershed algorithm, preprocessing techniques such as **Gaussian Filtering** can be used to remove noise, and **Morphological Operations** can be applied to identify foreground and background markers. The watershed algorithm then separates the merged objects based on gradient information and predefined markers.

(d) Justification:

Marker-Controlled Watershed Segmentation is preferred because it accurately separates touching or merged objects by analyzing image gradients and marker locations rather than relying only on intensity values. It minimizes incorrect merging, preserves object boundaries, and provides better segmentation results than simple thresholding. The preprocessing steps further improve accuracy by reducing noise and preventing unnecessary segmentation errors.

(e) Structured Explanation:

When **under-segmentation** occurs, multiple objects are combined into a single region, making it difficult to identify them individually. The best solution is to apply **Gaussian Filtering** for noise reduction, followed by **Morphological Operations** to generate markers, and finally **Marker-Controlled Watershed Segmentation** to separate the merged objec

5. Boundary Refinement

Detected boundaries are rough and inaccurate.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in boundary detection.
(c) Apply a suitable refinement method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

In this scenario, the object boundaries detected after segmentation or edge detection are rough, irregular, and inaccurate. Instead of smooth and continuous outlines, the boundaries contain jagged edges, gaps, or small distortions. These inaccuracies reduce the quality of segmentation and affect the accuracy of object measurement, recognition, and analysis.

(b) Key Challenges in Boundary Detection:

The major challenges include image noise, blurred or weak edges, irregular boundary shapes, low contrast between the object and background, and discontinuous edge pixels. These problems lead to incorrect object boundaries, making segmentation less reliable and reducing the performance of subsequent image processing tasks.

(c) Suitable Refinement Method:

The most suitable method is Morphological Operations, particularly Morphological Closing followed by Boundary Smoothing. Closing fills small gaps and connects broken boundary segments, while smoothing removes jagged edges and produces a clean object outline. If higher accuracy is required, Active Contour (Snake) Model can also be used to refine the boundary.

(d) Justification:

Morphological Closing is preferred because it connects nearby boundary pixels, removes small holes, and produces continuous boundaries without significantly changing the object's shape. Boundary smoothing further improves the appearance by eliminating irregular edges. Compared with simply applying another edge detector, this method preserves important object details while creating accurate and smooth boundaries.

(e) Structured Explanation:

When detected boundaries are rough and inaccurate, the best approach is to first apply Morphological Closing to connect broken edges and fill small gaps. Next, Boundary

Smoothing (or Active Contour for complex images) is used to refine the object outline.

6. Frequency-Based Enhancement

An image requires enhancement using frequency domain methods.
(a) Interpret the scenario and explain frequency-based processing.
(b) Identify key issues in spatial domain methods.
(c) Apply an appropriate frequency domain technique.
(d) Justify your choice logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The image requires enhancement using the **frequency domain**, where the image is transformed using the **Fourier Transform (FFT)**. Low frequencies represent the overall image, while high frequencies represent edges and fine details.

(b) Key Issues in Spatial Domain Methods:

The major issues are **limited control over frequency components**, edge blurring during smoothing, and difficulty in removing periodic noise.

(c) Appropriate Frequency Domain Technique:

The most suitable technique is the **High-Pass Filter (HPF)**. The image is transformed using **FFT**, filtered with HPF, and converted back using **Inverse FFT (IFFT)**.

(d) Justification:

The **High-Pass Filter** enhances edges and fine details by emphasizing high-frequency components, providing better control than spatial domain sharpening methods.

(e) Structured Explanation:

For frequency-based enhancement, the image is processed using **FFT → High-Pass Filter → IFFT**. This method enhances edges, improves image sharpness, and preserves important details effectively.

7. Mixed Noise Scenario

An image contains both Gaussian and impulse noise.

(a) Interpret the scenario and explain challenges.
(b) Identify key issues affecting filtering.
(c) Apply a suitable combined approach.
(d) Justify your method logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The image contains **both Gaussian noise and impulse (salt-and-pepper) noise**, making image enhancement more challenging than dealing with a single type of noise. **Gaussian noise** appears as random intensity variations throughout the image, while **impulse noise** appears as isolated black and white pixels. Since these two noise types have different characteristics, using only one filtering technique cannot effectively remove both without affecting image quality.

(b) Key Issues Affecting Filtering:

The major challenges include **random intensity fluctuations caused by Gaussian noise**, **black-and-white impulse pixels**, blurred edges after excessive smoothing, loss of fine details, and reduced accuracy in feature extraction and image analysis. Choosing an inappropriate filter may remove one type of noise while leaving the other or may blur important image features.

(c) Suitable Combined Approach:

The most suitable approach is to first apply a **Median Filter** to remove **impulse (salt-and-pepper) noise**, followed by a **Gaussian Filter** to reduce the remaining **Gaussian noise**. If edge preservation is important, a **Bilateral Filter** can be used instead of the Gaussian Filter because it smooths the image while preserving edges.

(d) Justification:

This sequence is effective because the **Median Filter** specifically removes impulse noise without blurring object edges. Once the impulse noise is removed, the **Gaussian Filter** smooths the random Gaussian noise, producing a cleaner image with fewer intensity variations. Applying the Gaussian Filter first could spread impulse noise to neighboring pixels, making it more difficult to remove. Therefore, using the filters in this order provides better noise reduction while preserving important image details.

(e) Structured Explanation:

When an image contains **both Gaussian and impulse noise**, a single filter is not sufficient because each noise type requires a different removal technique. The recommended pipeline is **Median Filter → Gaussian Filter** (or **Median Filter → Bilateral Filter** for better edge preservation). This combined approach removes salt-and-pepper noise first, then reduces Gaussian noise, resulting in a smooth image with preserved edges, improved clarity, and better performance in subsequent image processing tasks such as segmentation and feature extraction.

8. Edge Preservation Filtering

A filtering method is required to remove noise but preserve edges.

(a) Interpret the scenario and explain the requirement.
(b) Identify key issues in filtering.
(c) Apply a suitable technique.
(d) Justify your approach logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The image contains noise that reduces its quality, but the object edges must be preserved because they contain important information for image analysis. The requirement is to remove noise without blurring or weakening the object boundaries. This is essential in applications such as medical imaging, industrial inspection, and object recognition, where accurate edge information is critical.

(b) Key Issues in Filtering:

The major challenges include **image noise**, edge blurring caused by conventional smoothing filters, loss of fine details, reduced contrast, and inaccurate feature extraction. Using a filter that removes noise aggressively may also remove important edge information.

(c) Suitable Technique:

The most suitable technique is the **Bilateral Filter**. This filter removes noise by considering both the spatial distance and intensity difference between neighboring pixels, allowing smoothing within regions while preserving sharp edges.

(d) Justification:

The **Bilateral Filter** is preferred because it effectively reduces noise while maintaining edge sharpness. Unlike the Mean or Gaussian filters, which blur edges during smoothing, the Bilateral Filter preserves object boundaries by avoiding smoothing across significant intensity changes. This results in better image quality and more accurate edge-based analysis.

(e) Structured Explanation:

When noise must be removed without affecting important edges, the **Bilateral Filter** is the best choice. It smooths homogeneous regions while preserving sharp boundaries, producing a clean image with clear edges. This improves image quality and supports accurate segmentation, feature extraction, and object detection.

9. Region Growing Failure

Region growing fails due to incorrect seed selection.

(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in region growing.
(c) Apply a suitable correction.
(d) Justify your method logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

In **Region Growing Segmentation**, the algorithm starts from a **seed point** and adds neighboring pixels with similar properties. If the seed point is selected incorrectly, the region grows into the wrong area or fails to include the entire object, resulting in inaccurate segmentation.

(b) Key Challenges in Region Growing:

The major challenges include **incorrect seed selection**, image noise, intensity variations, weak object boundaries, and similar intensity values between the object and background. These factors can cause incomplete segmentation or leakage into neighboring regions.

(c) Suitable Correction:

The most suitable correction is to use **Automatic Seed Selection** along with **preprocessing using Gaussian Filtering** to remove noise. Morphological operations can also be applied to improve the seed locations before performing Region Growing.

(d) Justification:

Automatic Seed Selection chooses seed points from homogeneous regions instead of relying on manual selection, reducing segmentation errors. Gaussian filtering removes noise, while morphological operations improve the object structure, allowing the region to grow accurately without leaking into unwanted areas.

(e) Structured Explanation:

When **Region Growing** fails due to incorrect seed selection, the best approach is to **first apply Gaussian Filtering**, then use **Automatic Seed Selection**, followed by **Region Growing Segmentation**. This sequence reduces noise, selects accurate seed points, prevents region leakage, and produces reliable and accurate segmentation.

10. Industrial Inspection Enhancement

An industrial system requires detecting defects in noisy images.
(a) Interpret the scenario and define requirements.
(b) Identify key issues in detection.
(c) Apply an appropriate enhancement and segmentation pipeline.
(d) Justify your sequence logically.
(e) Present your explanation clearly.

ANSWER
(a) Interpretation of the Scenario:

An industrial inspection system must detect defects such as **cracks, scratches, dents, or surface flaws** in noisy images. The requirement is to **remove noise, enhance image quality, and accurately segment defects** for reliable inspection.

(b) Key Issues:

The major issues are **image noise**, blurred edges, low contrast, false defect detection, and difficulty in separating defects from the background.

(c) Appropriate Enhancement and Segmentation Pipeline:

Median/Gaussian Filter → Contrast Enhancement (CLAHE) → Canny Edge Detection → Threshold-Based Segmentation (or Morphological Operations).

(d) Justification:

The filter first removes noise, **CLAHE** improves image contrast, **Canny Edge Detection** accurately detects defect boundaries, and **Thresholding/Morphological Operations** separate the defects from the background. This sequence improves detection accuracy and minimizes false detections.

(e) Structured Explanation:
For industrial defect detection, the best pipeline is **Noise Removal → Contrast Enhancement → Edge Detection → Segmentation**. This sequence produces clear, well-defined defect regions, improves inspection accuracy, and enables reliable quality control.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand